Shweta Mittal

CASE 1: BLS

(i) Optimal purchasing plan and total cost (1225 GWh case)

	Madison	*Barton*	*Carroll*	*Story*	*Hardin*	*Franklin*	*Greene*	*Webster*
Prices ($k/GWh)	$49.50	$50.00	$61.00	$63.50	$66.50	$71.00	$72.50	$80.00
Active Developer	Yes	Yes	No	Yes	No	Yes	No	No
East / West	West	East	West	East	East	East	West	West
% Renewable	15%	16%	18%	20%	21%	22%	23%	25%
Capacity (GWh / year)	300	600	510	655	575	680	450	490
Electricity purchased (GWh)	55	600	0	20	100	0	450	0
Atleast 50% from active sources	62.5	0						
Atmost 650GWh from West	505	650						
Atmost 720GWh from East	720	720						
Atleast 19% from renewables	0	0						
Required GWh/year	1225							
total electricity purchased	1225							
Total cost of Electricity ($k)	73267.5							

Optimal electricity purchases (GWh)

- Madison: **55**
- Barton: **600**
- Carroll: **0**
- Story: **20**
- Hardin: **100**
- Franklin: **0**
- Greene: **450**
- Webster: **0**

Total cost

- Cost = **73,267.5 ($k) = $73.27 million**

Interpretation

- BLS buys heavily from **Barton (cheap + active)** and **Greene (renewable-heavy)**.
- Smaller allocations to **Madison, Story, Hardin** help satisfy:
  - ≥50% active supplier constraint

    - ≥19% renewable constraint
- Several suppliers (Carroll, Franklin, Webster) are **not used at all** because they are relatively expensive or less useful for constraints.

The optimal purchasing plan concentrates heavily on Barton and Greene, as they provide a strong balance of low cost and favorable characteristics for meeting both the "active developer" and renewable energy constraints. Additional smaller purchases from Madison, Story, and Hardin are used to ensure that BLS satisfies the minimum requirements (at least 50% from active suppliers and at least 19% from renewable sources) while also respecting east–west transmission limits and supplier capacities. More expensive or less strategically useful suppliers, such as Carroll, Franklin, and Webster, are excluded. Overall, the solution reflects a cost-minimizing strategy that carefully balances economic and environmental constraints.

(ii) Sensitivity to demand changes

Case A: Demand increases to 1250 GWh

	Madison	*Barton*	*Carroll*	*Story*	*Hardin*	*Franklin*	*Greene*	*Webster*
Prices ($k/GWh)	$49.50	$50.00	$61.00	$63.50	$66.50	$71.00	$72.50	$80.00
Active Developer	Yes	Yes	No	Yes	No	Yes	No	No
East / West	West	East	West	East	East	East	West	West
% Renewable	15%	16%	18%	20%	21%	22%	23%	25%
Capacity (GWh / year)	300	600	510	655	575	680	450	490
Electricity purchased (GWh)	72	600	0	0	120	0	450	8
Atleast 50% from active sources	47	0						
Atmost 650GWh from West	530	650						
Atmost 720GWh from East	720	720						
Atleast 19% from renewables	0	0						
Required GWh/year	1250							
total electricity purchased	1250							
Total cost of Electricity ($k)	74809							

New optimal purchases

- Madison: **72**
- Barton: **600**
- Carroll: **0**
- Story: **0**
- Hardin: **120**

- Franklin: **0**
- Greene: **450**
- Webster: **8**

Total cost

- **74,809 ($k) = $74.81 million**

Change vs. base

- Cost increases by ~ **$1.54M**
- Additional demand is met by:
  - Increasing **Madison and Hardin**
  - Introducing **Webster (8 GWh)** despite high cost
- Indicates **capacity/constraint limits are binding**, forcing use of more expensive suppliers.

The core structure of the solution remains similar, with Barton and Greene still supplying large portions of the demand. However, because key constraints such as supplier capacities and transmission limits become more binding, BLS must rely more on higher-cost suppliers, including introducing Webster and increasing purchases from others like Madison and Hardin. This results in a higher total cost, showing that additional demand is met at the margin using less cost-efficient options.

Case B: Demand decreases to 1200 GWh

	Madison	*Barton*	*Carroll*	*Story*	*Hardin*	*Franklin*	*Greene*	*Webster*
Prices ($k/GWh)	$49.50	$50.00	$61.00	$63.50	$66.50	$71.00	$72.50	$80.00
Active Developer	Yes	Yes	No	Yes	No	Yes	No	No
East / West	West	East	West	East	East	East	West	West
% Renewable	15%	16%	18%	20%	21%	22%	23%	25%
Capacity (GWh / year)	300	600	510	655	575	680	450	490
Electricity purchased (GWh)	30	600	0	120	0	0	450	0
Atleast 50% from active sources	150	0						
Atmost 650GWh from West	480	650						
Atmost 720GWh from East	720	720						
Atleast 19% from renewables	0	0						
Required GWh/year	1200							
total electricity purchased	1200							
Total cost of Electricity ($k)	71730							

New optimal purchases

- Madison: **30**
- Barton: **600**
- Carroll: **0**
- Story: **120**
- Hardin: **0**
- Franklin: **0**
- Greene: **450**
- Webster: **0**

Total cost

- **71730 ($k) = $71.73 million**

Change vs. base

- Cost decreases by ≈ **$1.54M**
- BLS shifts toward:
  - **Story (cheaper than Hardin)**
  - Eliminates **Hardin entirely**
- More flexibility allows substitution toward **lower-cost suppliers**

BLS gains more flexibility and can shift purchases toward relatively cheaper suppliers. In this case, higher-cost suppliers like Hardin are reduced or eliminated, and more cost-effective suppliers such as Story are used instead. The core suppliers (Barton and Greene) remain central, but the reduced need allows BLS to avoid expensive marginal purchases, leading to a lower overall cost. This demonstrates that when demand decreases, cost savings are achieved by dropping the least efficient suppliers from the mix.

Shweta Mittal

Case 2: Foreign Exchange

Rates				Buy		
		GBP	CAD	EUR	CHF	USD
	GBP	-	1.8587	1.3272	1.3097	1.4988
	CAD	0.5372	-	0.7135	0.7042	0.8057
Sell	EUR	0.7525	1.3999	-	0.9870	1.1282
	CHF	0.7631	1.4197	1.0131	-	1.1453
	USD	0.6664	1.2396	0.8847	0.8731	-

Transactions				Buy			Total
		GBP	CAD	EUR	CHF	USD	Sold
	GBP	0	0	0	0	0	0
	CAD	0	0	0	0	0	0
Sell	EUR	5.979879	0	0	0	0	5.9798793
	CHF	0	7.043792	0	0	0	7.04379224
	USD	0	0	0	0	0	0
Total bought		4.5	10	0	0	0	

Positions							
		Initial	Bought	Sold	Final		Required
	GBP	0	4.500	0	4.500		4.5
	CAD	0	10.000	0	10.000		10
	EUR	8	0	5.979879	2.020121		0
	CHF	10	0	7.043792	2.956208		0
	USD	1.2	0	0	1.2		0
					Final USD Value of Holdings		$21.666272
					Initial USD Value of Holdings		$21.678265
					Transaction Costs		$0.011992

To maximize the U.S. dollar value of final holdings, Loro should make two exchanges only: sell 5.979879 million EUR to purchase 4.5 million GBP, and sell 7.043792 million CHF to purchase 10 million CAD. No USD should be exchanged, and no other currency trades are needed. These transactions exactly satisfy the required final holdings in GBP and CAD while preserving the highest possible remaining portfolio value. After the exchanges, Loro ends with 2.020121 million EUR, 2.956208 million CHF, and 1.2 million USD in

addition to the required 4.5 million GBP and 10 million CAD. The initial USD value of holdings is 21.678265 million USD, while the final USD value is 21.666272 million USD, implying a loss of about 0.011993 million USD (about $11,993) due to bank spreads embedded in the exchange rates. This loss is unavoidable, but the Solver solution minimizes it.